

MYSQL ASSIGNMENT 2 -QUERYING DATA

CREATION OF DATABASE AND TABLES:

Created database named **employee**

Created three tables named **department _info**, **employees** and **location**.

```
120      -- 1. Departments Table
121  ● ○ CREATE TABLE Departments_Info (
122      department_id INT PRIMARY KEY,
123      department_name VARCHAR(100) NOT NULL UNIQUE
124  );
125
126      -- 2. Locations Table
127  ● ○ CREATE TABLE Locations (
128      location_id INT auto_increment primary KEY,
129      location_name VARCHAR(100) NOT NULL UNIQUE
130  );
131
132  ● ○ -- 3. Employees Table
133  ● ○ CREATE TABLE Employees (
134      employee_id INT PRIMARY KEY,
135      employee_name VARCHAR(100) NOT NULL,
136      gender VARCHAR(10) CHECK (gender IN ('M','F')),
137      age INT CHECK (age >= 18),
138      hire_date DATE NOT NULL,
139      designation VARCHAR(200),
140      salary DECIMAL(10,2),
141      department_id INT,
142      location_id INT,
143
144      FOREIGN KEY (department_id)
145          REFERENCES Departments_Info(department_id)
146          ON DELETE CASCADE
147          ON UPDATE CASCADE,
148      FOREIGN KEY (location_id)
149          REFERENCES Locations(location_id)
150          ON DELETE CASCADE
151          ON UPDATE CASCADE
152  );
```

INSERTION VALUE CODE:

These are the values of the tables

```
153 • INSERT INTO departments_info(department_id, department_name) VALUES
154     (1, 'Software Development'),
155     (2, 'Marketing'),
156     (3, 'Data Science'),
157     (4, 'Human Resources'),
158     (5, 'Product Management'),
159     (6, 'Content Creation'),
160     (7, 'Finance'),
161     (8, 'Design'),
162     (9, 'Research and Development'),
163     (10, 'Customer Support'),
164     (11, 'Business Development'),
165     (12, 'IT'),
166     (13, 'Operations');
167
168 • INSERT INTO locations (location_id,location_name) VALUES
169     (1,'Chennai'),
170     (2,'Bangalore'),
171     (3,'Hyderabad'),
172     (4,'Pune');
173
174 • INSERT INTO employees (employee_id, employee_name, gender, age, hire_date, designation, department_id, location_id, salary) VALUES
175     (5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),
176     (5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),
177     (5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),
178     (5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
179     (5005, 'Anvi Chaudhary', 'F', 25, '2015-09-11', 'Business Development Executive', 11, 1, 55000),
180     (5006, 'Dhruv Shetty', 'M', 28, '2015-11-20', 'UI Developer', 8, 2, 65000),
181     (5007, 'Anushka Singh', 'F', 32, '2016-01-15', 'Marketing Manager', 2, 3, 90000),
182     (5008, 'Diya Jha', 'F', 27, '2016-03-05', 'Graphic Designer', 8, 4, 70000),
183     (5009, 'Kiaan Desai', 'M', 30, '2016-05-20', 'Sales Executive', 11, 3, 55000),
184     (5010, 'Atharv Yadav', 'M', 29, '2016-07-10', 'Systems Administrator', 12, 4, 80000),
185     (5011, 'Saarvi Patel', 'F', 28, '2016-09-20', 'Marketing Analyst', 2, 1, 60000),
186     (5012, 'Mitesh Verma', 'F', 26, '2016-11-05', 'Operations Manager', 13, 2, 85000);
```

CLAUSE & OPERATORS:

1. DISTINCT VALUES:

A query to retrieve distinct salaries from the Employees table.

```
206      -- distinct salary
207 •    SELECT DISTINCT salary
208      FROM Employees;
209
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: ☐

salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

2. ALIAS (AS):

aliases for the "age" and "salary" columns as "Employee_Age" and "Employee Salary", respectively.

```
210      --      "Employee_Age" and "employee salary"
211 •    SELECT
212      AGE AS EMPLOYEE_AGE,
213      SALARY AS EMPLOYEE_SALARY
214      FROM EMPLOYEES;
215
216      -- employees salary greater than ₹50000 and hired before 2016-01-01.
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: ☐

EMPLOYEE_AGE	EMPLOYEE_SALARY
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	65000.00
32	90000.00
27	70000.00
30	55000.00
29	80000.00
28	60000.00
26	95000.00
33	75000.00
30	60000.00
27	85000.00
31	100000.00
28	80000.00
25	75000.00
29	55000.00
32	90000.00

3. WHERE CLAUSE & OPERATORS:

1.employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
216 -- employees salary greater than ₹50000 and hired before 2016-01-01.
217 • SELECT
218 EMPLOYEE_NAME
219 FROM EMPLOYEES
220 WHERE SALARY>50000 AND HIRE_DATE <'2016-01-01';
221
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
EMPLOYEE_NAME				
Vihaan Singh				
Reyansh Singh				
Kiara Malhotra				
Anvi Chaudhary				
Dhruv Shetty				

2. employee whose designation is missing and fill it with "Data Scientist".

```
222 -- SET NULL INTO DATA SCIENTIST
223 • UPDATE Employees
224 SET designation = 'Data Scientist'
225 WHERE designation IS NULL
226 AND employee_id > 0;
227
```

```
5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),
5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
5005, 'Anvi Chaudhary', 'F', 25, '2015-09-11', 'Business Development Executive', 11, 1, 55000),
```

5003	Aaradhya Iyer	F	26	2015-05-20	Customer Support Executive	45000.00	10	2
5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	70000.00	8	3
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	55000.00	11	1
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	65000.00	8	2

SORTING AND GROUPING DATA:

1. ORDER BY:

employees sorted by department ID in ascending order and salary in descending order.

```
228 -- employees sorted by department ID in ascending order and salary in descending order.
229 • SELECT employee_id,
230         employee_name,
231         SALARY
232 FROM Employees
233 ORDER BY department_id ASC,
234         salary DESC;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
employee_id	employee_name	SALARY		
5020	Aarav Nair	90000.00		
5007	Anushka Singh	90000.00		
5023	Anika Paul	70000.00		
5011	Saanvi Patel	60000.00		
5025	Avni Iyengar	100000.00		
5026	Vivaan Singh	75000.00		
5001	Vihaan Singh	60000.00		
5016	Zoya Khan	100000.00		
5022	Aadhya Iyengar	60000.00		
5019	Ishika Patel	55000.00		
5024	Aryan Shetty	95000.00		
5030	Sara Iyer	90000.00		
5027	Ananya Paul	60000.00		
5028	Anaya Kapoor	60000.00		
5015	Ishaan Kumar	85000.00		
5021	Advik Kapoor	85000.00		
5004	Kiara Malhotra	70000.00		

2. LIMIT

the first 5 employees hired in the year 2018.

```
236 -- Display the first 5 employees hired in the year 2018.
237 • SELECT employee_id,
238         employee_name,
239         hire_date
240 FROM Employees
241 WHERE YEAR(hire_date) = 2018
242 ORDER BY hire_date ASC
243 LIMIT 5;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell C
employee_id	employee_name	hire_date		
5019	Ishika Patel	2018-01-15		
5020	Aarav Nair	2018-03-05		
5021	Advik Kapoor	2018-05-20		
5022	Aadhya Iyengar	2018-07-10		
5023	Anika Paul	2018-09-20		
NULL	NULL	NULL		

3. AGGREGATE FUNCTIONS:

the sum of all salaries in the Finance department.

```
45 -- Calculate the sum of all salaries in the Finance department.
46 • SELECT
47     SUM(SALARY) AS TOTAL_SALARY
48     FROM EMPLOYEES
49     WHERE DESIGNATION='Finance Analyst';
50
51
```

TOTAL_SALARY
85000.00

Find the minimum age among all employees.

```
252 -- Find the minimum age among all employees.
253 • SELECT
254     MIN(AGE) AS MINIMUM_AGE
255     FROM EMPLOYEES;
```

MINIMUM_AGE
25

4. GROUP BY:

the maximum salary for each location.

```
259 -- List the maximum salary for each location.
260 • SELECT
261     LOCATION_ID,
262     MAX(SALARY) AS MAXIMUM_SALARY
263     FROM EMPLOYEES
264     GROUP BY LOCATION_ID;
```

LOCATION_ID	MAXIMUM_SALARY
1	95000.00
2	95000.00
3	90000.00
4	100000.00

average salary for each designation containing the word 'Analyst'.

```
264      -- average salary for each designation containing the word 'Analyst'.
265 •    SELECT designation,
266         AVG(salary) AS average_salary
267     FROM Employees
268     WHERE designation LIKE '%Analyst%'
269     GROUP BY designation;
270
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
designation	average_salary			
Data Analyst	60000.000000			
Marketing Analyst	60000.000000			
Supply Chain Analyst	60000.000000			
Financial Analyst	85000.000000			
Finance Analyst	85000.000000			
Business Analyst	75000.000000			
Quality Assurance Analyst	80000.000000			

5. HAVING:

departments with less than 3 employees.

```
271      -- Find departments with less than 3 employees.
272 •    SELECT department_id,
273         COUNT(employee_id) AS employee_count
274     FROM Employees
275     GROUP BY department_id
276     HAVING COUNT(employee_id) < 3;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
department_id	employee_count			
1	1			
5	2			
6	2			
7	2			
9	1			
11	2			
13	1			

locations with female employees whose average age is below 30.

```
278 -- Find locations with female employees whose average age is below 30.
279 • SELECT location_id,
280         AVG(age) AS avg_age
281 FROM Employees
282 WHERE gender = 'F'
283 GROUP BY location_id
284 HAVING AVG(age) < 30;
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

location_id	avg_age
1	26.7500
2	27.3333
4	29.2000

JOINS:

1. INNER JOIN:

employee names, their designations, and department names where employees are assigned to a department.

```
286 -- List employee names, their designations, and department names
287 • SELECT
288     E.EMPLOYEE_NAME,
289     E.DESIGNATION,
290     D.DEPARTMENT_NAME
291 FROM EMPLOYEES AS E
292 INNER JOIN
293 DEPARTMENTS_INFO AS D
294 ON E.DEPARTMENT_ID=D.DEPARTMENT_ID;
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

EMPLOYEE_NAME	DESIGNATION	DEPARTMENT_NAME
Anvi Chaudhary	Business Development Executive	Business Development
Kiaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support
Arnav Rao	Customer Success Manager	Customer Support
Vihaan Mohan	Supply Chain Analyst	Customer Support
Vihaan Singh	Data Analyst	Data Science
Avni Iyengar	Data Scientist	Data Science
Vivaan Singh	Business Analyst	Data Science
Kiara Malhotra	Data Scientist	Design
Dhruv Shetty	UI Developer	Design
Diya Jha	Graphic Designer	Design
Ishaan Kumar	Financial Analyst	Finance
Advik Kapoor	Finance Analyst	Finance
Zoya Khan	Legal Counsel	Human Resources

Result 22 x

2. LEFT JOIN:

all departments along with the total number of employees in each department, including departments with no employees.

```
300     D.DEPARTMENT_NAME,  
301     count(E.employee_id) as toyal_employees  
302     from DEPARTMENTS_INFO AS D  
303     left join  
304     EMPLOYEES AS E  
305     ON E.DEPARTMENT_ID=D.DEPARTMENT_ID  
306     group by department_name;  
307
```

DEPARTMENT_NAME	toyal_employees
Business Development	2
Content Creation	2
Customer Support	3
Data Science	3
Design	3
Finance	2
Human Resources	3
IT	4
Marketing	3
Operations	1
Product Management	2
Research and Develo...	1
Software Development	1

3. RIGHT JOIN:

all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```
106     -- EMPLOYEE AND LOCATION USING RIGHT JOIN  
107     SELECT  
108         L.location_name,  
109         E.employee_name  
110     FROM Employees AS E  
111     RIGHT JOIN Locations AS L  
112     ON E.location_id = L.location_id;
```

location_name	employee_name
Bangalore	Aaradhya Iyer
Bangalore	Dhruv Shetty
Bangalore	Myra Verma
Bangalore	Vihaan Mohan
Bangalore	Kabir Nair
Bangalore	Anika Paul
Bangalore	Vivaan Singh
Bangalore	Arjun Kumar
Chennai	Reyansh Singh
Chennai	Anvi Chaudhary
Chennai	Saanvi Patel
Chennai	Ishaan Kumar
Chennai	Aarav Nair
Chennai	Aryan Shetty
Chennai	Anaya Kapoor
Chennai	Sara Iyer

result 25 x

output

4. CROSS JOIN:

all possible combinations of departments and locations.

```
314      -- CROSS JOIN LOCATION AND DEPARTMENT NAME
315 •    SELECT
316      D.department_name,
317      L.location_name
318  FROM Departments_Info AS D
319  CROSS JOIN Locations AS L;
320
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	department_name	location_name
▶	Business Development	Pune
	Business Development	Hyderabad
	Business Development	Chennai
	Business Development	Bangalore
	Content Creation	Pune
	Content Creation	Hyderabad
	Content Creation	Chennai
	Content Creation	Bangalore
	Customer Support	Pune
	Customer Support	Hyderabad
	Customer Support	Chennai
	Customer Support	Bangalore
	Data Science	Pune
	Data Science	Hyderabad
	Data Science	Chennai
	Data Science	Bangalore

Result 26 x

5. SELF JOIN:

pairs of employees working in the same department, excluding self-pairs.

```
322      -- Show pairs of employees working in the same department, excluding self-pairs.
323 •    SELECT e1.employee_id AS emp1_id,
324      e1.employee_name AS emp1_name,
325      e2.employee_id AS emp2_id,
326      e2.employee_name AS emp2_name,
327      e1.department_id
328  FROM employees as e1
329  JOIN employees e2
330  ON e1.department_id = e2.department_id
331  AND e1.employee_id < e2.employee_id
332  ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	emp1_id	emp1_name	emp2_id	emp2_name	department_id
▶	5001	Vihaan Singh	5025	Avni Iyengar	3
	5001	Vihaan Singh	5026	Vivaan Singh	3
	5002	Reyansh Singh	5010	Atharv Yadav	12
	5002	Reyansh Singh	5017	Kabir Nair	12
	5002	Reyansh Singh	5029	Arjun Kumar	12
	5003	Aaradhya Iyer	5013	Arnav Rao	10
	5003	Aaradhya Iyer	5014	Vihaan Mohan	10
	5004	Kiara Malhotra	5006	Dhruv Shetty	8
	5004	Kiara Malhotra	5008	Diya Jha	8
	5005	Anvi Chaudhary	5009	Kiaan Desai	11
	5006	Dhruv Shetty	5008	Diya Jha	8
	5007	Anushka Singh	5011	Saanvi Patel	2
	5007	Anushka Singh	5023	Anika Paul	2
	5010	Atharv Yadav	5017	Kabir Nair	12
	5010	Atharv Yadav	5029	Arjun Kumar	12
	5011	Saanvi Patel	5023	Anika Paul	2

Result 3 x

WINDOWS FUNCTION:

window function query to rank employees by salary using rank ().

```
136 • SELECT
137     employee_id,
138     employee_name,
139     salary,
140     RANK() OVER (ORDER BY salary DESC) AS SalaryRank
141 FROM Employees;
```

employee_id	employee_name	salary	SalaryRank
5016	Zoya Khan	100000.00	1
5025	Avni Iyengar	100000.00	1
5012	Myra Verma	95000.00	3
5024	Aryan Shetty	95000.00	3
5007	Anushka Singh	90000.00	5
5020	Aarav Nair	90000.00	5
5030	Sara Iyer	90000.00	5
5015	Ishaan Kumar	85000.00	8
5021	Advik Kapoor	85000.00	8
5002	Reyansh Singh	80000.00	10
5010	Atharv Yadav	80000.00	10
5017	Kabir Nair	80000.00	10
5029	Arjun Kumar	80000.00	10
5013	Arnav Rao	75000.00	14
5018	Ishan Mishra	75000.00	14
5026	Vivaan Singh	75000.00	14

Result 4 x

window function query to rank employees by salary within each department using DENSE_RANK ()

```
15 • SELECT
16     employee_id,
17     employee_name,
18     department_id,
19     salary,
20     DENSE_RANK() OVER (PARTITION BY department_id ORDER BY salary DESC) AS DeptSalaryRank
21 FROM Employees;
```

result Grid | Filter Rows: | Export: | Wrap Cell Content: |

employee_id	employee_name	department_id	salary	DeptSalaryRank
5020	Aarav Nair	1	90000.00	1
5007	Anushka Singh	2	90000.00	1
5023	Anika Paul	2	70000.00	2
5011	Saanvi Patel	2	60000.00	3
5025	Avni Iyengar	3	100000.00	1
5026	Vivaan Singh	3	75000.00	2
5001	Vihaan Singh	3	60000.00	3
5016	Zoya Khan	4	100000.00	1
5022	Aadhya Iyengar	4	60000.00	2
5019	Ishika Patel	4	55000.00	3
5024	Aryan Shetty	5	95000.00	1
5030	Sara Iyer	5	90000.00	2
5027	Ananya Paul	6	60000.00	1
5028	Anaya Kapoor	6	60000.00	1
5015	Ishaan Kumar	7	85000.00	1
5021	Advik Kapoor	7	85000.00	1

result 5 x

window function query, Running total salary by department

```
53 -- window function query, Running total salary by department
54
55 • SELECT
56     employee_id,
57     employee_name,
58     department_id,
59     salary,
60     SUM(salary) OVER (PARTITION BY department_id ORDER BY salary DESC) AS RunningTotalSalary
61 FROM Employees;
62
```

result Grid | Filter Rows: | Export: | Wrap Cell Content: |

employee_id	employee_name	department_id	salary	RunningTotalSalary
5020	Aarav Nair	1	90000.00	90000.00
5007	Anushka Singh	2	90000.00	90000.00
5023	Anika Paul	2	70000.00	160000.00
5011	Saanvi Patel	2	60000.00	220000.00
5025	Avni Iyengar	3	100000.00	100000.00
5026	Vivaan Singh	3	75000.00	175000.00
5001	Vihaan Singh	3	60000.00	235000.00
5016	Zoya Khan	4	100000.00	100000.00
5022	Aadhya Iyengar	4	60000.00	160000.00
5019	Ishika Patel	4	55000.00	215000.00
5024	Aryan Shetty	5	95000.00	95000.00
5030	Sara Iyer	5	90000.00	185000.00
5027	Ananya Paul	6	60000.00	120000.00
5028	Anaya Kapoor	6	60000.00	120000.00
5015	Ishaan Kumar	7	85000.00	170000.00
5021	Advik Kapoor	7	85000.00	170000.00

result 6 x